A Discrete Causal Lattice Foundation for Physics: The Octahedral $\mathcal{A} = 1$ Framework

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Abstract

[STUB] The abstract will summarize the A=1 framework applied to a three-dimensional octahedral causal lattice $\mathcal{T}_\diamond^3$, in which every particle carries an independent tick counter and physics emerges from the combinatorial scheduling of those counters. Gravity arises as a clock-density gradient rather than spacetime curvature. A falsifiable prediction concerning discrete path-count deviations from the continuous $1/r^2$ law is derived and proposed for experimental verification.

Contents

1 Introduction **3**

2 The Octahedral Substrate ($\mathcal{T}_\diamond^3$) **3**

3 Causal Sessions and the Phase Rotor **3**

4 The Tick Scheduler **3**

5 Phase Propagation and the Unity Constraint **3**

6 Emergent Kinematics: Momentum and Force in the $\mathcal{T}_{3d}$ Manifold **3**

7 Emergent Kinematics on the $\mathcal{T}_\diamond^3$ Lattice **3**

7.1 The Bipartite Velocity 4

7.2 Phase Gradient as Momentum 4

7.3 Acceleration and Force 6

8 Gravity as Clock Density **6**

9 Clock Fluid Dynamics and the Continuum Limit **6**

10 The Vacuum Twist: Gravity, Electromagnetism, and the Unified Phase Field **7**

11 Interference and the Huygens Lantern **7**

12 The Observer as Clock **7**

13 Path Counting and Falsifiable Predictions **7**

14 Lattice Harmonics and Quantization **7**

15 The Hydrogen Spectrum from Lattice Geometry **7**

16 System Audit and Traceability Matrix **7**

17 Conclusion **7**

1 Introduction

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2 The Octahedral Substrate ($\mathcal{T}_\diamond^3$)

[**STUB**]

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[**STUB**]

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[**STUB**]

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[**STUB**]

6 Emergent Kinematics: Momentum and Force in the $\mathcal{T}_{3d}$ Manifold

7 Emergent Kinematics on the $\mathcal{T}_\diamond^3$ Lattice

Momentum, inertia, and force are not postulated in the $\mathcal{A} = 1$ framework. They emerge from the geometry of phase interference across the bipartite sublattice structure. This section derives the kinematic relations from the Dirac tick rule and connects each result to the corresponding numerical verification in the experiment suite.

7.1 The Bipartite Velocity

A massive Dirac fermion [1] on $\mathcal{T}_\diamond^3$ carries two spinor components: ψ_R on the RGB sublattice (even ticks) and ψ_L on the CMY sublattice (odd ticks). The $\mathcal{A} = 1$ constraint requires

$$\sum_{\mathbf{x}} \left(|\psi_R(\mathbf{x})|^2 + |\psi_L(\mathbf{x})|^2 \right) = 1. \tag{1}$$

Over one complete two-tick cycle the kinetic fraction $p_{\text{hop}} = \cos^2(\omega/2)$ is distributed across the six basis vectors, split equally between the three RGB directions ($\mathbf{V}_{1\text{–}3}$, even tick) and the three CMY directions ($-\mathbf{V}_{1\text{–}3}$, odd tick):

$$1 = p_{\text{stay}} + P_{\text{fwd}} + P_{\text{bwd}}, \qquad p_{\text{stay}} = \sin^2\tfrac{\omega}{2}, \quad P_{\text{fwd}} + P_{\text{bwd}} = \cos^2\tfrac{\omega}{2}. \tag{2}$$

Here $P_{\text{fwd}} = \sum_{i=1}^{3} p(\mathbf{V}_i)$ is the total probability flowing along the RGB sublattice and $P_{\text{bwd}} = \sum_{i=1}^{3} p(-\mathbf{V}_i)$ along CMY. The lattice speed limit $c = 1$ node/tick gives the net velocity

$$v = c\,(P_{\text{fwd}} - P_{\text{bwd}}). \tag{3}$$

For a packet at rest the two sublattices are populated symmetrically, $P_{\text{fwd}} = P_{\text{bwd}} = \frac{1}{2}\cos^2(\omega/2)$, so $v = 0$ and the amplitude oscillates between ψ_R and ψ_L without net displacement — the lattice realisation of Zitterbewegung [2].

7.2 Phase Gradient as Momentum

A uniform spatial phase gradient with advance θ per lattice edge breaks the RGB/CMY symmetry. In the implementation this gradient appears as the per-direction phase advance

$$\delta_p(\mathbf{V}_i, \mathbf{x}) = \arg\psi(\mathbf{x} + \mathbf{V}_i) - \arg\psi(\mathbf{x}), \tag{4}$$

which is the quantity `delta_p` computed in `CausalSession._kinetic_hop`. Only positive advances contribute to the hop weight, so amplitude flows preferentially in the direction of increasing phase — the lattice realisation of the de Broglie relation [3].

To derive $v(\theta)$ analytically, consider a packet with mass parameter ω (the instruction frequency) subject to a gradient θ directed along the net RGB axis. On the even (RGB) tick ψ_R hops forward: the phase gradient $+\theta$ reduces the effective mismatch from ω to $\omega - \theta$, increasing the kinetic weight. On the odd (CMY) tick ψ_L hops backward: the reversed

gradient adds $+\theta$ to the mismatch, decreasing the kinetic weight:

$$P_{\text{fwd}} = \tfrac{1}{2}\cos^2\tfrac{\omega-\theta}{2}, \qquad (\psi_R \text{ hop, even tick}) \tag{5}$$

$$P_{\text{bwd}} = \tfrac{1}{2}\cos^2\tfrac{\omega+\theta}{2}. \qquad (\psi_L \text{ hop, odd tick}) \tag{6}$$

Substituting into Eq. (3) and applying $\cos^2\alpha - \cos^2\beta = -\sin(\alpha+\beta)\sin(\alpha-\beta)$ yields

$$v = \tfrac{c}{2}\,\sin(\omega)\,\sin(\theta). \tag{7}$$

Table 1 illustrates how p_{stay}, m_{inertial}, and the resulting velocity vary across representative values of ω: In the macroscopic limit $\theta \ll 1$ (gradient small compared to 2π), $\sin\theta \approx \theta$.

ω	$p_{\text{stay}} = \sin^2(\omega/2)$	$m_{\text{inertial}} = \sin(\omega)/2$	v/c at $\theta = 0.1$
0	0	0	1 (photon limit)
$\pi/4$	0.146	0.354	0.035
$\pi/2$	0.500	0.500	0.050
π	1.000	0	0 (frozen, max mass)

Table 1: Selected values of the two mass-related quantities and the resulting lattice velocity at a small phase gradient $\theta = 0.1$. The photon row ($\omega = 0$) is handled by the massless limit (Section **??**); the other rows follow Eq. (7).

Identifying the lattice momentum $p \equiv m\,v/c$ and the effective mass $m \equiv \sin(\omega)/2$[1] recovers the classical kinematic relation

$$p = m\,\theta \cdot c, \tag{8}$$

where θ is the phase advance per lattice edge — the discrete analogue of the wavenumber k. Momentum is thus strictly the cross-term between the particle's internal mass-phase ω and the spatial phase gradient of the bipartite vacuum.

Massless limit. Equation (7) gives $v = 0$ when $\omega = 0$, which is not the photon behaviour. The massless case is qualitatively distinct: with $\omega = 0$ the residence probability vanishes, $p_{\text{stay}} = 0$, and the amplitude hops entirely via the directional weighting of Eq. (4). The hop is strictly one-way along the gradient ($\delta_p > 0$ only), so $P_{\text{bwd}} = 0$ and $v = c$ regardless of θ. The two-sublattice derivation above applies to massive particles ($\omega > 0$); the photon is the $\omega \to 0$ limit treated separately in Section **??**.

[1] The inertial mass $m_{\text{inertial}} = \sin(\omega)/2$ is *not* the same as the residence probability $p_{\text{stay}} = \sin^2(\omega/2)$. They coincide only at $\omega = \pi/2$, the unique frequency at which the particle spends exactly half its time at rest. At $\omega = \pi$ (maximum residence) the particle is frozen, $m_{\text{inertial}} = 0$, while $p_{\text{stay}} = 1$.

Numerical verification. Experiment `exp_01_inertia.py` initialises a Gaussian wavepacket ($\sigma = 3$, $\omega = 0.2$) with momentum $\mathbf{k} = (k, k, k)$, $k = 0.3/\sqrt{3}$, corresponding to a phase advance $\theta = \mathbf{k}\cdot\mathbf{V}_1 = 0.3\sqrt{3} \approx 0.520$ rad per hop along the $\mathbf{V}_1 = (1, 1, 1)$ direction. Equation (7) predicts

$$v_x = \tfrac{1}{2}\sin(0.2)\sin(0.3\sqrt{3}) \approx 0.0493 \text{ nodes/tick}.$$

The measured x-component fit over 12 ticks gives $v_x = 0.0464$ nodes/tick — a 6% discrepancy attributable to finite packet width and the short run length. The heavy particle ($\omega = 0.5$, same $\mathbf{k}$) drifts measurably slower, confirming that $\sin(\omega)$ governs inertial resistance: a larger phase cost per hop reduces the net kinetic probability imbalance.

7.3 Acceleration and Force

Acceleration requires a time-varying gradient. Taking the discrete time derivative of Eq. (7) with ω held constant:

$$a = \frac{\Delta v}{\Delta t} = \frac{c}{2}\,\sin(\omega)\,\cos(\theta)\,\frac{\Delta\theta}{\Delta t}. \tag{9}$$

For small gradients ($\cos\theta \approx 1$) this reduces to the lattice analogue of Newton's second law [4]:

$$a = \frac{c}{2m}\,F_\theta, \qquad F_\theta \equiv \frac{\Delta\theta}{\Delta t}. \tag{10}$$

The quantity F_θ — the rate at which the spatial phase gradient changes per tick — is the lattice definition of force. A gravitational or electromagnetic field acts by twisting the phase of the vacuum over time, unbalancing P_{fwd} and P_{bwd} and continuously converting p_{stay} into directed kinetic probability. The mechanism by which the Coulomb potential produces F_θ in the hydrogen orbital is analysed in Section **??**.

8 Gravity as Clock Density

[STUB]

9 Clock Fluid Dynamics and the Continuum Limit

[STUB – see notes/clock_fluid_dynamics.md]

10 The Vacuum Twist: Gravity, Electromagnetism, and the Unified Phase Field

[STUB – see notes/vacuum_twist_field_equations.md]

11 Interference and the Huygens Lantern

[STUB]

12 The Observer as Clock

[STUB]

13 Path Counting and Falsifiable Predictions

[STUB]

14 Lattice Harmonics and Quantization

[STUB – see notes/lattice_harmonics.md and exp_09_harmonics.py]

15 The Hydrogen Spectrum from Lattice Geometry

[RESULT: r_2/r_1 = 4.42, E_2/E_1 = 0.229. Bohr spectrum confirmed at 8-10%. Spontaneous n=2 to n=1 decay observed at tick 340. Both results are emergent.]

16 System Audit and Traceability Matrix

17 Conclusion

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Table 2: System Audit: Mapping $\mathcal{T}_\diamond^3$ Protocols to Experiments

Physical Phenomenon	**Lattice Logic**	**Manifold Limit**	**Experiment**	**Status**
Speed of Light Limit	Topological Adjacency ($v_{max} = 1$)	Planck Ratio ($c = l_P/t_P$)	exp_00_causal_cone.py	PASS
Inertial Persistence	Phase Gradient ($\nabla\phi$)	Newton's First Law	exp_01_inertia.py	PASS
Gravitational Pull	Clock Density Gradient	Schwarzschild Metric	exp_02_gravity_clock_density.py	PASS
Quantum Interference	Discrete Path Summation	Schrödinger Equation	exp_03_interference.py	PASS
Wave Function Collapse	Observer Clock Addition	Density Matrix Decay	exp_04_decoherence.py	PASS
Measurement Problem	Tick Sequence Entanglement	Decoherence Theory	exp_05_observer_clock.py	PASS
$1/r^2$ Correction	Discrete Path Count Deviation	Gaussian CLT Limit	exp_06_path_counting.py	PASS
Time Dilation Quantum	Minimum Clock Increment	Schwarzschild Time Dilation	exp_06_path_counting.py	PASS
Clock Density Conservation	Continuity Equation from tick()	Einstein Field Equations	exp_07_clock_conservation.py	PASS
Gravity vs EM Twist & div(ϕ) vs curl(ϕ) & Maxwell + Einstein & exp_08_vacuum_twist.py & STUB				
Zitterbewegung Spectrum	Frequency scan, stability peaks	Mass quantization	exp_09_harmonics.py	PASS
Orbital Resonance	Stable orbits in clock-density well	Bohr spectrum	exp_09_harmonics.py	PART
Photon Dispersion	Group velocity vs wavenumber k	Brillouin zone boundary	exp_09_harmonics.py	PASS
Temporal Locking	Frequency drift toward rationals	Charge quantization	exp_09_harmonics.py	PART
Hydrogen Spectrum	Coulomb well orbital resonance	Bohr $\sim -1/n^2$	exp_10_hydrogen_spectrum.py	CONF
Spontaneous Emission	Zitterbewegung phase damping	n=2→1 decay	exp_10_hydrogen_spectrum.py	CONF

References

[1] Paul A. M. Dirac. The quantum theory of the electron. *Proceedings of the Royal Society A*, 117(778):610–624, 1928.

[2] Erwin Schrödinger. About the force-free motion in relativistic quantum mechanics. *Sitzungsberichte der Preussischen Akademie der Wissenschaften*, pages 418–428, 1930. Original derivation of Zitterbewegung.

[3] Louis de Broglie. Recherches sur la théorie des quanta. *Annales de Physique*, 3:22–128, 1925. Wave–particle duality and the matter wavelength $\lambda = h/p$.

[4] Isaac Newton. *Philosophiæ Naturalis Principia Mathematica*. Royal Society, London, 1687.